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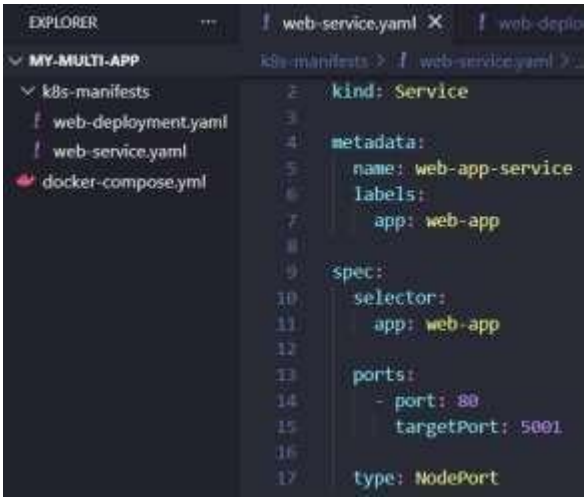
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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 13

Subject:	DevOps Lab
TCODE:	TE7950
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Branch	CSE, Batch (2023-27), L3
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23rd July, 2026
Title of Assignment:	Kubernetes Service & Networking
Theory:	A Kubernetes Service is an abstraction that provides a stable network endpoint for accessing one or more Pods in a Kubernetes cluster. Since Pods are temporary and their IP addresses may change, a Service uses labels and selectors to route incoming traffic to the appropriate Pods. A NodePort Service exposes the application outside the cluster by allocating a port on each cluster node, allowing external access to the application. During local development, `kubectl port-forward` can also be used to securely forward traffic from a local machine to a Kubernetes Service for testing and verification. This approach simplifies communication between applications and ensures reliable access to containerized workloads. **Commands Used** • Created the Kubernetes Deployment: `kubectl apply -f web-deployment.yaml` • Verified that the Pods were running: `kubectl get pods` • Created the Kubernetes Service: `kubectl apply -f`

	`web-service.yaml` • Displayed the available Kubernetes Services: `kubectl get service` • Displayed detailed information about the Service: `kubectl describe service web-app-service` • Displayed the Service endpoints: `kubectl get endpoints` • Displayed the running Pods with labels: `kubectl get pods --show-labels` • Displayed the application logs: `kubectl logs web-app-deployment-<pod-name>` • Forwarded the Kubernetes Service to the local machine: `kubectl port-forward service/web-app-service 8085:80` • Deleted the Kubernetes Service: `kubectl delete -f web-service.yaml` • Deleted the Kubernetes Deployment: `kubectl delete -f web-deployment.yaml`
Implementation:	**Step 1: Create the Kubernetes Service YAML**  Created Kubernetes Service Manifest — Created the `web-service.yaml` file defining a Kubernetes NodePort Service to expose the Flask web application running in the Kubernetes Pods. **Step 2: Apply the Service**

```
PS C:\Users\riavi\my-multi-app\k8s-manifests> kubectl apply -f web-deployment.yaml
deployment.apps/web-app-deployment created
PS C:\Users\riavi\my-multi-app\k8s-manifests> kubectl get pods
NAME                                 READY   STATUS    RESTARTS   AGE
web-app-deployment-848b75c97-c6r58   1/1     Running   0          20s
web-app-deployment-848b75c97-jqxbb   1/1     Running   0          20s
PS C:\Users\riavi\my-multi-app\k8s-manifests> kubectl apply -f web-service.yaml
service/web-app-service created
```

Applied Kubernetes Service — Applied the Kubernetes Service manifest using `kubectl apply`, creating a NodePort Service for the deployed Flask application.

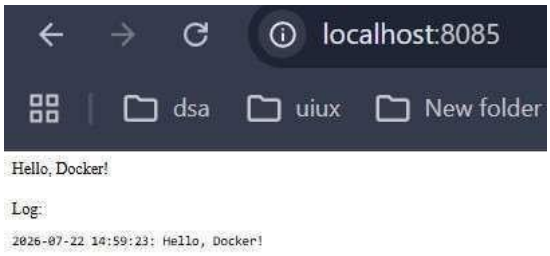
Step 3: Find the Service Information

```
PS C:\Users\riavi\my-multi-app\k8s-manifests> kubectl get service
NAME              TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
kubernetes        ClusterIP   10.96.0.1       <none>        443/TCP          25m
web-app-service   NodePort    10.96.139.63    <none>        80:30649/TCP     59s
```

Verified Kubernetes Service — Verified the Kubernetes NodePort Service and identified the assigned NodePort (30649) used to expose the Flask web application outside the Kubernetes cluster.

Step 4: Access the Application

```
PS C:\Users\riavi\my-multi-app\k8s-manifests> kubectl port-forward service/web-app-service 8085:8
Forwarding from 127.0.0.1:8085 -> 5001
Forwarding from [::1]:8085 -> 5001
Handling connection for 8085
Handling connection for 8085
```



Accessed Application Through Kubernetes Service — Used `kubectl port-forward` to forward the Kubernetes Service to the local machine and successfully accessed the Flask web application at `http://localhost:8085`, verifying that it was running correctly.

Step 5: Clean Up

```
service "web-app-service" deleted from default namespace
```

	Deleted Kubernetes Service — Removed the Kubernetes Service using the `kubectl delete` command, cleaning up the networking resources created for the application.

Subject Faculty Name: Prof. Arijit Dutta